

Gender Inequality in Deliberative Participation *

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Abstract:

Normative theorists assert that deliberation rightly constructed can promote equality of voice, while critics argue that many deliberative settings actually encourage inequality of voice, particularly for disadvantaged groups such as women. Using experimental data, we explore the conditions under which gender equality of verbal participation occurs. We document the pervasive gender gap in voice and in efficacy, confirming the critics' concern. However, we find that the structure of deliberation can exacerbate or erase this inequality: the interaction between the group's decision rule and its gender composition has a substantial effect on voice and efficacy. Women are most disadvantaged when they are either a numerical minority under majority rule or a large majority under unanimous rule. They reach equality when they are a large majority under majority rule. In addition, both men and women maximize their individual participation in gender-homogenous, "enclave" deliberations. We find that voice substantially mediates the effects of decision rule and gender composition on perceptions of efficacy. Deliberative design must content with pervasive inequality, but can avoid it by fitting the right institutional procedures to the social context of the situation.

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For some time now, normative theorists have urged more citizen deliberation (Barber 1984; Bohman 1997; Chambers 1996, 2003; Fishkin 1995; Gutmann and Thompson 1996, 2004; Habermas 1989, 1996; Macedo 1999). One of the key dimensions of desirable deliberation as articulated by many theorists is equality of participation. Gutmann and Thompson state, for example, that deliberation rightly constructed will “diminish the discriminatory effects of class, race, and gender inequalities” (2004, p. 48, 50). In the article that in many ways sparked the deliberative turn in political theory, Joshua Cohen writes that in ideal deliberation the “existing distribution of power and resources does not shape [deliberators’] chances to contribute to deliberation, nor does that distribution play an authoritative role in their deliberation” (p. 74).

One of the main worries about implementing deliberation, however, is that it will not achieve this equality. Instead, deliberation may *create* difficulties for groups that are disadvantaged in terms of status or power, specifically through the mechanism of voice (Young 1996; 2001; Sanders 1997; Fraser 1992; Williams 2000).

Pragmatically, the task for those who structure actual deliberations is to minimize the deleterious effects of class, race, or gender inequalities on discussion. Government units at the local, state and federal levels have been increasingly turning to group discussions for input into policy-making or for conflict resolution. Small group discussions are also ubiquitous in civic life, in voluntary organizations, workplaces and educational settings (Jacobs, Cook, and Delli Carpini 2009; Fung 2007; Ryfe 2005; Cramer Walsh 2007; Merelman et al. 1998; Macedo et al. 2005). How women and minorities can achieve voice and representation in these settings has become an increasingly important policy question (e.g., Grutter v. Bollinger 2003). The National Health Planning and Resources Development Act, for example, required citizens serving on boards to be “broadly representative” of social identities (Mansbridge 1999, p. 634). Some states have passed laws requiring gender balance (Iowa Legal Code 2010). But the conditions that allow disadvantaged social groups to achieve equal voice are little understood.

Our goal is to explore the extent to which inequalities exist in deliberating groups while assessing the conditions that might exacerbate or mitigate them. We focus in particular on the amount of talk as a measure of participation during deliberation. We are not aware of any quantitative studies of citizen deliberation that analyze the behavior of deliberators (see Cramer Walsh 2007 and Rosenberg 2007 for qualitative analyses). Yet, as the theorists cited above argue, and as Mansbridge (1983) has underscored, speech is a crucial type of political act in a democracy, and all the more so in deliberative settings. In order to characterize the extent and conditions that give rise to inequality in deliberation, we need to examine the volume of voice and the patterns of silence.

We examine speech as a form of symbolic political or civic participation that may reflect and contribute to the sense of political efficacy and the broad civic capacity of a disadvantaged social group (Kathlene 1994). We will not examine whether speaking increases the chances that the group implements the speaker’s preferences. That is a worthy question, but one that requires full treatment and therefore lies beyond the scope of this paper. Rather, our study addresses the rationales offered by advocates of deliberation and the critics’ concerns, both of which are less concerned with whose

preferences are implemented then with deliberation's potential to enlarge or undermine democratic competence and republican capacities (Gastil et al 2008; Luskin and Fishkin 1999; see also Pateman 1970). This also follows the lead of studies of women's voting and political participation such as Burns et al (2001), which also treat political participation in as a civic act. Therefore, we treat speech as a type of civic act and assess inequalities in the level of speech participation. As a second measure of individuals' civic capacity we also assess perceptions of efficacy.

Using an experimental design we test several hypotheses about the conditions that increase or decrease equality. We vary two features of deliberation that have been implicated by prior studies: the group's gender composition and the group's decision rule. We find that these factors do matter, but in a departure from prior studies, we find that the effects of one depend on the other.

Literature Review

A. Group Gender Composition¹

Why and how might the number of women shape the level of gender equality in participation?² One prominent explanation, which we will call the Gender Role hypothesis, is that in groups where women discuss politics with men, women tend to defer to the assumed expertise and displayed confidence of men. Men tend to be seen as more competent in tasks viewed as masculine -- those involving agency and instrumentality (Ridgeway 1982). Men tend to be perceived as more competent and to enjoy a higher status than women in discussions of what are perceived to be masculine subjects (Ridgeway and Smith Lovin 1999). Politics has long been viewed as a "man's game" (Burns et al. 2001). As a result, women may defer to men when interacting with them in mixed-gender groups, but this deference may evaporate when men are absent, in all-female groups (Ridgeway 1982; Aries 1998). Consistent with this Gender Role hypothesis, there is evidence that in mixed-gender interactions, women tend to speak less, feel less confident, and exercise less influence than men during political or legal discussions (Eagly 1987; Giles et al. 1987; Huckfeldt and Sprague 1995, chapter 10; Bowers et al. 2001; Hastie et al. 1983; Strodtbeck et al. 1957; Croson and Gneezy 2009). We may thus expect women's participation and efficacy to be higher in all-female than in gender-mixed deliberating groups. Indeed, this rationale underlies some theorists' advocacy of protected "enclave" discussions (Karpowitz, Raphael and Hammond 2009).

In addition, gender role theory expects that numerical status affects women's participation and efficacy in the group. Johnson and Schulman (1989) conducted a study of efficacy ratings of

¹ Individual-level differences across genders are often small and inconsistent in comparison to large variances within gender (Sapiro 2003; Aries 1998). However, these differences can become large and consequential when amplified by group forces (Sapiro 2003).

² Scholarship on group composition consistently affirms the notion that gender composition shapes group and individual decisions (Aries 1996; Johnson and Schulman 1989; see Karpowitz and Mendelberg 2007 for a review). For example, gender composition affects judicial outcomes even if individual gender does not affect preferences, and even after controlling on ideology and other factors (Massie, Johnson, and Gubala (2002); Farhang and Wawro (2004); Peresie (2005); Mendelberg (2008)). These effects extend beyond the legal context (Aries 1976; Karpowitz et al 2008; Smith-Lovin and Brody 1989). It is possible that gender composition effects weaken without a group decision (Farrar et al 2009, footnote 7).

participants in group discussions by fellow participants. They found that while both men and women are rated lower when they are in a numerical minority vs. a majority, women incur a greater disadvantage.³ Therefore, we might expect that the gender gap in participation might grow in groups where women are a minority, and attenuate in groups with more women, where there are fewer active, confident participants to which the women would defer (Smith-Lovin and Brody 1989; Aries 1998; Johnson 1994; Piliavin and Martin 1978).

Women may speak less in discussions involving many men than in discussions with many women for an additional reason -- gendered norms of interaction appear to vary with the gender composition of the setting and may facilitate or hinder women's participation. In settings with many men, the interaction tends to take on more stereotypically masculine characteristics of individual assertion, agency, competition and dominance, and in settings with many women, people tend to interact in a more stereotypically feminine style that emphasizes cooperation, intimacy and the inclusion of all participants (Aries 1976; Ellis 1982; McCarrick, Manderscheid and Silbergeld 1981; Miller 1985; see Mendelberg and Karpowitz 2007 for a literature review). Men tend to adopt a more assertive, even aggressive, or less intimate style of speech in settings where they are predominant (Aries 1976; Smith Lovin and Brody 1989). Conversely, levels of self-disclosure during group discussion increase as the number of women in the group increases (Dindia and Allen 1992). Women may experience a greater sense of confidence in predominantly-female settings with their more stereotypically feminine norms of interaction, and more discomfort in settings with predominantly-male norms. This hypothesis receives support in studies of political settings – for example, from Kathlene's (1994) study of the Colorado state legislature. She found that predominantly male committees use a competitive, aggressive communication style and that this speaking behavior inhibits women's participation more than it does men's. In sum, in keeping with gender role theories, this literature predicts that women participate less than men when they are in predominantly male groups and increase their participation as the number of women increases.

The Gender Role hypothesis also makes a specific prediction about the effects of highly imbalanced gender compositions. As already implied by the above discussion, men may out-participate women to the greatest extent in groups composed of only one woman (female "token" groups, in the terminology of this literature). But gender role theory further suggests that women will be especially disadvantaged when they are the lone members of their group relative to men who are the lone members of their group. That is, token women's participation and efficacy will be lower than token men's. We refer to this as the Token hypothesis.

Gender role theory also produces the Rooster hypothesis which subsumes the Token hypothesis but goes a step further and argues that male tokens derive benefits not only relative to the women in their group but also relative to men in other groups. The Rooster hypothesis receives

³ They estimated the effects of randomly assigned gender compositions (ranging from 0 to 4 women) on participant ratings of other members' leadership following the group discussion. Among those rated as high in leadership, ratings of leadership decreased more for women than they did for men as the proportion of women or men declined, respectively. While high-leadership men as well as women are negatively affected by a decrease in their same gender proportions, the effects on ratings of men are of smaller magnitude, apply in more limited compositions (only when the man is the single male), and still leave men in a better position than high-leadership women in token groups and in gender-balanced groups.

support from studies finding that while female tokens in male occupations are disadvantaged, male tokens in female occupations are advantaged (Folger and Merrill 1985). There is some support for this proposition in small group lab experiments that vary gender composition and assess individual influence (Craig and Sherif 1986; Johnson and Schulman 1989). Perhaps, then, minority status advantages men relative to women in their groups, to men in other groups, and to minority status women in other groups.

Contradicting the Token and Rooster hypotheses, Kanter argues that small minority status affects men and women similarly because it affects the salience of the individual's gender. We call this the Equal Disadvantage hypothesis. When either men or women are very few in number, their gender becomes more salient and they are judged in more negative stereotypical terms (Kanter 1977a; 1977b; 1977c). We might thus predict that the disadvantages of token status are the same for women and men -- both will be less likely to talk and to be perceived as efficacious than the other members of their own group and of same-gender members of other group compositions.

The final hypothesis argues that men may be more negatively affected by numerical minority status than women are because they are less familiar with it. According to this Violated Entitlement hypothesis, when men are in a small minority, their expectation that they be more influential than women is violated and their sense of confidence may lessen. Women, on the other hand, may be used to being less influential than men and as a result of this low expectation may not be as affected by their token status (Chatman and O'Reilly 2004; van Knippenberg and Schippers 2007). We might therefore hypothesize that men will find their token status unusual and negative and speak less and be perceived as less efficacious in such a setting relative not only to the women in their group and to men in other groups, but most relevant, to token women.

B. Institutional Context and Decision Rule

While scholarship in sociology and psychology provides predictions about the effects of gender enity and of minority status, it does not attend much to decision rule. The contribution of political science to this area comes from the notion that institutions and their rules and procedures matter (Mansbridge 1983).

Previous work suggests that a unanimous decision rule can create group norms that enhance consensus. In line with this expectation, a seminal study of mock juries reports that people told to decide with unanimous rule were more likely to shift their views during discussion than people assigned to majority rule juries (Hastie et al. 1983). Unanimous rule can increase a sense that the decision was legitimate and appropriate (Kameda 1991; Kaplan and Miller 1987; Nemeth 1977). Group consensus generated through talk can also lead to increased cooperative behavior (Bouas and Komorita 1996). Given these findings (see Mendelberg 2002 for more detail), we might hypothesize that unanimous decision rule leads to consensus-oriented norms that protect numerical minorities and reduce gender inequality in speaking behavior and efficacy.

However, the literature also offers the opposite view. Consensus pressures can silence participants and increase rejection of the views of numerical minorities, ((Davis, Bray and Holt 1977; Davis, Kameda, Parks, Stasson and Zimmerman 1989; Davis, Stasson, Ono and Zimmerman 1988; Miller et al. 1987; Penrod and Hastie 1980; Mansbridge 1983; Karpowitz and Mansbridge 2005).

Falk and Falk (1981) find that majority decision rule may counteract inequities of influence more effectively than unanimous rule. We might, therefore, expect that unanimous rule will not protect numerical minorities and might cause less equal talk and efficacy.

We test these two contradictory hypotheses about decision rule against each other and add the interaction of their effect with the effects of gender composition. If unanimous rule creates consensus norms that aid numerical minorities, then unanimous rule may help women in the minority to overcome the disadvantage they may experience in predominantly male groups. If unanimous rule hurts numerical minorities and majority rule helps them, then we should see that majority rule erases the disadvantage that women experience in predominantly male groups. Regardless of the direction of effects, we propose that the effects of gender composition are contingent on the institutional rules under which groups operate.

Data and Methods

We designed the study to meet the following requirements: 1) generate a sufficient number of groups in various gender compositions to create adequate variance in the independent variable;⁴ 2) test for the predicted interactive effects of gender composition with decision rule, 3) use random assignment to create exogenous gender composition and decision rule variables so as to gauge their unbiased effects, 4) measure the level of verbal participation of individuals and match it to their individual characteristics, including their gender.⁵

The experiment followed the basic procedure of a study by Frohlich and colleagues (Frohlich, Oppenheimer, and Eavey 1987; Frohlich and Oppenheimer 1990; 1992). In their experiment participants were told that they would be doing tasks to earn money; that the money they actually receive would be based on a group decision about redistribution, with the decision based on either unanimous or majority rule⁶; but that prior to group deliberation, they would not be told the nature of the work they would be doing. This was meant to simulate the Rawlsian veil of ignorance designed to prompt people to consider principles of justice. In addition the investigators instructed participants to make a decision that would not only apply concretely and immediately to themselves and their group, but also would be applied hypothetically to society at large, in order to generalize beyond the lab situation to the decisions people make about redistribution in politics. We replicate and extend this procedure since it allows us to test the effects of assigned decision rule and gender composition on deliberation.

Between August 2007 and February of 2009, students and community members were recruited, randomly assigned to one of the conditions, and brought together for two hours. After

⁴ We stratified by gender, since randomly assigning individuals to a group without stratifying by gender composition will generate a balanced gender mix in most groups and this would produce insufficient variance in gender composition of the groups.

⁵ Each individual was recorded on a separate track, along with a master track that included all participants. A software application (Verbal Behavior Analysis Software v1.3) is publicly available for research purposes at [redacted]. A technical appendix is available at: [redacted]

⁶ A control condition of no deliberation (excluded from this paper) was included for purposes other than speech analysis.

participants filled out a pre-treatment questionnaire, we instructed participants to conduct a “full and open discussion” and to choose the “most just” principle of redistribution. The only requirement was that they deliberate for at least five minutes; on average, the groups we analyze spent just over 25 minutes (standard deviation=11) in discussion.⁷ All instructions were exactly the same across conditions. After deliberation, participants performed several rounds of work -- a spelling correction task -- and earned money according to their performance and their chosen distribution scheme. At the end they responded to a series of questions on attitudes and beliefs and debriefed.⁸

We use a 6 X 2 between subjects experimental design. We randomly assigned individuals to one of 6 gender compositions (that is, to a group that ranged from 0 to 5 women) and one of two decision rule conditions consisting of unanimous rule or majority rule.⁹ Randomization checks and propensity score analyses find that individuals were assigned by a random process and groups are equivalent on relevant covariates.¹⁰

In what follows, we report findings for 470 individuals in 94 groups. Table 1 presents the number of groups and individuals in each experimental condition examined here.

Table 1 about here

The experiment was conducted at two different sites – a small town on the Atlantic coast and a medium-sized city in the Mountain West. In all analyses, we control on site since subjects are assigned randomly within but not across sites. As is common in controlled experiments of this kind, our goal was not a nationally representative sample but one with reasonable variance, and the sample did in fact vary in relevant variables such as SES and political characteristics (see online appendix).¹¹

Results

A. Participation

We begin with the extent to which the conversation is equally divided across the various group members. We divide the number of seconds each individual spoke by the total number of seconds when any group member was speaking. One virtue of this measure is that it controls for differences in the total talk time of a group so that the measure is not skewed by groups that happened to have long or short discussion times. We label this variable *Percent Talk*. Its empirical

⁷ The range is between 7 and 60 minutes. Participants agreed by unanimous vote to end deliberation. Voting occurred by secret ballot.

⁸ See [redacted] for the effects on other outcomes.

⁹ Gender composition was randomly assigned to a day and subjects who attended on that day were assigned to that day's randomly assigned composition. Randomization of decision rule was achieved by the roll of dice prior to the arrival of participants on the day of each experimental session. Additional information about the research design of the experiment is available at: [redacted]

¹⁰ Demographic characteristics such as education, income, age, partisanship, and student status had no significant relationship with the group's gender composition and decision rule. We performed three tests on each set of propensity scores: a two-sample t-test, a Wilcoxon-Mann-Whitney test and a Kolmogorov-Smirnov test, and they all yielded null results. Full details are available from the authors.

¹¹ Based on our reading of relevant literatures, we expected that individual race and group racial composition would introduce powerful interactions with our explanatory variables; since we would require a much larger number of groups to analyze these, we ran the study only on white non-Hispanic subjects. A study on racial composition is in the works. In addition, convenience recruitment generated disproportionate number of young participants.

range runs from .007 to .584 (mean = .20, st. dev. = .11). Across all groups, the mean level of *Percent Talk* for men is .215 (st. dev. = .12), while the mean for women is .185 (st. dev. = .10). A simple difference of means tests shows that this gender difference of 3 percentage points is statistically significant ($p < .01$). Men generally take up more of the conversation than women.

The critical question is how this gender inequality is affected by group context. Panels A and B of Figure 1 present separate predicted *Percent Talk* averages for men and women under the various experimental conditions. In discussions with perfect equality of participation, each participant in a group of five deliberators would contribute 20 percent of the total discussion.¹² Stars in the figure denote statistically significant differences from that line of equality. The predicted values are generated from a random effects regression that controls for differences across experimental sites and includes a variable for female, for the unanimous decision rule, and an interaction between female and unanimous decision rule.¹³ In addition, we are interested in the effect of the individual's gender because this variable measures a person's likely response to the dynamics of the group's interaction. But an individual's gender probably also measures preferences and views about equal redistribution. In order to estimate the effects of gender that are not due simply to differences between the preferences and opinions of men and women on the issue at hand, we control for outcome preferences by including a variable tapping whether or not the individual's pre-deliberation preferences about the most just principle of redistribution matched the group's eventual decision, and including a measure of pre-deliberation level of egalitarianism (see the supplementary appendix for all questions wordings and response options and for descriptive statistics on all variables used in the analyses). The regression was run separately for each gender composition, which allows us to capture possible non-linearities in the effects of gender composition. (See the Appendix, Table A1 for the details of regression models from which values are predicted.) For the analyses that follow, other tests, including difference of means tests and OLS models with cluster robust standard errors, produce similar results. In addition, other specifications, such as models that exclude controls for egalitarianism and pre-deliberation preferences, produce essentially identical patterns to those reported below.

Figure 1 about here

Panel A of Figure 1 shows high but contingent inequality of verbal participation. Opposite to the Gender Role hypothesis, in groups that decide by unanimous decision rule, equality occurs when women are in the minority, but significant inequality of speaking behavior emerges when women are the majority. These results imply that in a 30-minute conversation between four women and one man, the average man in a four-female group would speak for approximately 8.5 minutes, while the remaining four women would average less than 5.5 minutes of talk time each.¹⁴ Men speak more than

¹²In gender homogenous groups, then, perfect gender equality of talk will always be present by definition because the group's total talk time is divided across 5 group members of the same gender.

¹³Hausman tests indicate that random effects models are preferred to fixed effects with these data. Breusch-Pagan Lagrange multiplier tests also lead to the same conclusion. Predicted values are estimated by holding the variables of interest constant and integrating across all other variables in the model. In other words, these are average predicted values, sometimes called predictive marginals.

¹⁴This pattern is not the result of outliers skewing the averages. If we use median Percent Talk, the difference between males and females is nearly 3 minutes in a 30-minute discussion.

half again as much as the average woman in these groups. Similarly, in three-female groups, the average woman's percentage is about 70 percent of the average man's. Despite the fact that women make up 60 percent of the group, they collectively take barely half of the speaking time of the group's deliberation.

Panel B shows that gender inequality is also high and even more widespread under majority decision rule. Women in the minority hold the floor for much less and men for much more than their equal share. For example, women who were the token female in their group spoke for about 60 percent of the average man's percentage of floor time. In a 30-minute deliberation, these women speak for less than 4 minutes. It is only when women outnumber men 4-1 that women reach the 20 percent line of equality.

These gender effects are not driven by participants' preferences about the issues under consideration. The models from which these figures are generated include controls for participants' pre-deliberation commitment to equality and the match between their preferences and the group outcome. Though there are differences between men and women in their preferences about income equality – women in our sample are slightly more committed to equality and feel more warmly toward the poor than men – the presence of controls for these preferences in our models does not change the basic finding. It appears that gender – not issue preferences – is driving patterns of verbal behavior.

Figure 2 about here

So far the results suggest that under majority rule, the more women, the more equality, while under unanimous rule, the more women, the less equality. To test more directly for differences across gender compositions, and to show more clearly the interaction of gender composition with decision rule, we present Figure 2. Each point represents the difference between men's average percent talk and women's average percent talk, given the gender composition condition. With unanimous rule, minority women reach equality with men but majority women fall behind men. The gender gap is not statistically distinguishable from zero in the one- or two-female conditions, but is large and statistically significant in the three- or four-female condition. In the majority rule condition, the pattern is the opposite – women under-participate men in every composition except when women are a super majority, though the gap is somewhat smaller and does not quite reach statistical significance in the two-female condition.¹⁵

Most important for our argument that group composition and decision rule interact, the figure also confirms the interaction by showing that decision rule moves the gender gap in opposite directions. The gender gap differs across rule by a statistically significant magnitude where women are a super-majority, and approaches significance where women are tokens.

¹⁵ The gender gap in the 2-female condition falls short of standard levels of statistical significance in a two-tailed test ($p=0.15$). The gender gaps in groups with 1 and 3 women differ from the 4-female gap by $p<0.04$ (chi-square=4.26 and 4.39, respectively). The difference between 4-female gender gap and the gap in the groups with 2 women again falls short of standard levels of significance ($p<.15$, chi-square=2.06). These differences are computed from a random effects model in which all gender composition conditions are included.

Figure 2 also provides evidence about token status. Specifically, we find no support for the Rooster hypothesis in either the unanimous or the majority rule conditions. In the unanimous condition, the male advantage within the group is not significantly greater for groups with one male than for groups with two males.¹⁶ In majority-rule groups, the difference in gender gaps between the one-and two-male groups is statistically significant (chi-square=4.39, $p=0.04$), but the two-male groups have the significant advantage, not the groups with token males. The Violated Entitlement hypothesis is also disconfirmed because the gender gap never reverses – it simply recedes to zero.¹⁷

Figure 3 about here

Figure 3 provides further tests of the token status hypotheses by comparing token men against token women. The Equal Disadvantage hypothesis holds that males and females are similarly disadvantaged by being tokens and so their talk time will be the same. But Figure 3 suggests otherwise: the verbal participation of token men always exceeds that of token women. The difference approaches significance at the .10 level in the unanimous condition; in the majority condition it falls somewhat short of statistical significance; and pooling the two rule conditions, the difference is significant ($p=0.04$). There is thus only weak support at best for the Equal Disadvantage hypothesis.

The Gender Role prediction that token women do worse than token men fits the results better than the Rooster and Equal Disadvantage hypotheses; however, Figure 3 underscores that the Gender Role hypothesis still does not fully account for the difference between male and female tokens, because it neglects decision rule. Again, the interaction with decision rule matters. First, token females appear to do better with unanimous rule than they do with majority rule (the difference is 7.5 percentage points).¹⁸ In fact, unanimous rule protects female tokens sufficiently that they do about as well as token males do in the majority condition.¹⁹ Second, token males also do better with unanimous than majority rule (by 10.3 percentage points, $p < .05$). Both of these findings support the hypothesis that the unanimous rule protects numerical minorities.

However, the findings so far show that the advantage of the unanimous condition for token females comes at the cost of the disadvantage it creates for female majorities. Hence, unanimous rule creates equality for token females but exacerbates the advantage of token males. In fact, the greatest gender inequality between tokens occurs in the unanimous condition. Thus, unanimous rule protects numerical minorities, as the decision rule hypothesis predicts; but it interacts with gender, protecting men more than women, and thus fails to protect socially disadvantaged groups when taken as a whole.

¹⁶ A Wald test shows that the difference between the estimated value of the gender gap in the one-male groups and the gap in the two-male groups is not statistically significant (chi-square = 0.28, $p=0.60$).

¹⁷ The gender gap is negative in the four-female majority decision-rule condition, but that negative gap is not statistically distinguishable from zero.

¹⁸ The difference does not reach standard levels of statistical significance, (chi-square=1.93, $p=0.19$).

¹⁹ The p value of the difference between token females in the unanimous condition and token males in the majority condition is 0.91, chi-square=0.01.

A similar story emerges with an alternative measure of deliberative participation, *Percent Speaking Turns*.²⁰ We again find gender equality in unanimous groups with small numbers of females and male advantage when women become the majority. In the majority rule groups, on the other hand, we again find that equality of speaking behavior emerges only when women heavily outnumber men. (See Appendix Table A2 for details.)

The final measure of deliberative participation that we employ is the total time each participant held the floor.²¹ *Talk Time* is related to the *Percent Talk* measure (and was used to help construct that measure) but is also conceptually distinct. *Talk Time* has to do with the ability of an individual to fully articulate his or her thoughts, to voice his or her opinions in a more complete and whole way. In this sense it measures the level of individual self-expression. Moreover, the *Percent Talk* and *Percent Speaking Turns* measures do not clearly tell us whether men or women change their verbal behavior across conditions, since an individual's percentage inherently depends on others' behavior. One's *Talk Time*, by contrast, is much less dependent on the talk time of others. Finally, *Talk Time* also allows us to include comparisons with gender homogenous groups and thus test another prediction by the gender role hypothesis – that women do best in their own “enclaves”, away from the forces that produce gender inequality. The mean level of *Talk Time* for all participants is 3.85 minutes (st. dev. = 3.2), and ranges from a minimum of 5.25 seconds to a maximum of more than 16 minutes.

Figure 4 about here

Figure 4 shows predicted individual *Talk Time* in each experimental condition, with the same set of control variables as above.²² Consistent with the Gender Role hypothesis, female participants talked longest when surrounded by other women. In no gender composition is female talk time higher than in the gender homogenous groups. This finding holds across both the unanimous and majority decision rules. For women, gender-segregated groups are, it appears, conducive to individual self-expression and to greater verbal participation no matter what the decision rule. For men, estimated talk times were slightly higher (though the differences are not statistically significant) in the all-male groups deciding by majority rule, but much lower in the all-male groups deciding unanimously, compared to the mixed gender conditions.

Given this finding, as well as the theoretical importance of homogenous female groups for gender role theory, we use the standard set by homogenous groups as a comparison point for talk in mixed-gender groups. Carrots in the figure indicate a significant difference from the estimate for the

²⁰ This individual-level measure divides the number of times each participant took a speaking turn by the total number of speaking turns in the group.

²¹ Individual-level measures of talk time include both individual- and group-level variation. In this sense, the *Percent Talk* measure may be somewhat preferable for comparison across conditions, as it purges those group-level differences in total talk time.

²² Because we are primarily interested in the changing behavior of each gender across the different gender composition conditions, predicted values for these graphs were generated from a random-effects regression in which we examine men and women separately. This model includes controls for decision rule and gender composition condition.

gender homogenous group.²³ Panel A shows that in groups deciding by unanimous rule, male talk is significantly higher in groups with three or four females than it is in all male groups. Male talk times increase as they become the gender minority. Talk times for women, by contrast, differ less consistently between the mixed-gender and gender homogenous groups. Men more than women are the ones reacting to gender composition in the unanimous condition.²⁴ This finding sheds light on the mechanism for the inequality we found in our earlier analysis. Under unanimous rule, gender inequality grows when men talk more and, to a smaller extent, when women talk less than they do in gender homogenous groups.

Once again, however, as the rule-composition interaction hypothesis predicts, the pattern is very different for groups deciding by majority rule. Now it is women, not men, who are more sensitive to the gender composition condition. As Panel B shows, total talk times for men remain roughly constant across the various gender composition conditions. By contrast, women's talk times in 1-, 2-, and 3-female groups are significantly lower than in all-female groups. Only when women are the super-majority in the group do female talk times approach the levels found in gender homogenous groups. Thus, the unanimous decision rule condition seems to help women participate almost as fully in interaction with men as they do in all-female contexts, while the majority decision rule condition causes women to be much more quiescent until they strongly outnumber men in the group.

B. Efficacy

We turn next to the question of whether the participants' efficacy in the discussion was also affected. After the group discussion, we presented each participant with the following statement to measure their perceived self-efficacy: "My opinions were influential in shaping the group discussion and final decision." Participants responded to a 5-point scale ranging from strongly disagree to strongly agree. In addition, we also asked each group member to indicate the person who was "most influential" in the group's discussion and decisions. Participants were restricted to choosing only one member of their group. Because individuals were allowed to vote for themselves, their own votes are eliminated from the count.²⁵ For each group, then, we construct a tally of the number of votes each individual received, a measure that runs from 0 to 4. Just over 60 percent of the sample received 0 votes.²⁶ These two variables thus take us beyond individual-level subjective impressions of one's performance to form a group-level sense of a deliberator's deliberative efficacy and authority.

Figure 5 about here

Figure 5 shows the effects of deliberative participation on deliberative efficacy. The figure includes two panels: the number of group votes received from others (Panel A) and self-ratings of influence in Panel B. The lines for men and women only extend as far as the maximum value found

²³ Raw talk times are subject to greater variance because of the differences in the length of each group's discussion. For this reason, we use a one-tailed test when assessing statistical significance.

²⁴ The estimated values are slightly lower, but except for groups with three-females, the differences between the mixed-gender and gender homogenous groups are not statistically significant.

²⁵ Results reported below are essentially identical if individuals' own votes are included in the dependent variable.

²⁶ Among women, this number rises to nearly 70 percent.

in the data for each gender. The figures derive from Tables A4 and A5 in the Appendix (which uses negative binomial regression for others' perceptions, since the dependent variable skews toward 0).²⁷ The models include controls for pre-deliberation confidence in speaking ability (see Appendix for wording) and whether or not the individual's pre-deliberation preferences matched the group's eventual decision.²⁸ Results are similar when a host of additional controls are included in the models, including income, political preferences, and other demographic characteristics. As with the previous analyses, the patterns are the same if we use the percent of speaking turns in the group instead of *Percent Talk*.

Panel A of Figure 5 shows that participants who held the floor for a greater percentage of the group's deliberation were more likely to be seen as influential by the other members of the group. This is true for both men and women, though men generally tended to receive more votes than women.²⁹ Figure 5 thus demonstrates that for both men and women, active use of voice translates into greater authority in the group.

Participation also affected each individual's judgment of his or her own influence. As with the measure of group votes, women are generally less efficacious than men when it comes to self-assessments. Even so, those who spoke more were also more likely to feel efficacious. The predicted probability of a participant strongly agreeing that his or her voice was influential increases by 55 percentage points for men and 25 points for women as *Percent Talk* moves from its empirical minimum to maximum for each gender. These magnitudes are noteworthy given that we have controlled on the person's pre-deliberation sense of confidence in their speaking ability and for the actual group outcomes.

Figure 6

So far we have only indirect support for the notion that the interaction of decision rule and group composition shapes others' perception of a deliberator's efficacy. Panels A and B of Figure 6 provide direct support.³⁰ Women often have little efficacy, even approaching 0 votes (under majority rule with token women); they experience a pervasive efficacy deficit relative to men; and they do so in the same circumstances that produce the gender gap in speech participation -- with unanimous rule and many women, and under majority rule with few women. In other words, it appears that the experimental conditions affected group dynamics in ways that influenced both speaking behavior and post-deliberation evaluations of group influence.³¹ The remaining question is whether speaking behavior mediates those group conditions.

To complete the mediation analysis, Panels C and D of Figure 6 show the effect of the conditions on others' rating of influence, but this time including %Talk as a predictor alongside the

²⁷ In Panels A and B of the figure, predicted probabilities were constrained at the limit of possible votes in the group.

²⁸ Results are similar when controls for confidence in speaking ability and pre-deliberation preferences are excluded.

²⁹ Gender differences in the relationship between Percent Talk and votes received are not quite statistically significant in Panel A ($p < .11$). On average across all groups, men tended to receive about one-third more votes than women ($p < 0.01$).

³⁰ These figures based on predictive margins from a negative binomial regression with cluster robust standard errors. See Appendix Table A6 for the full model.

³¹ The results are similar when we replace the number of votes received with a binary variable for receiving the most votes in the group.

same predictors used previously (see Table A7 in the appendix). The results are consistent with the notion that speech participation mediates the effect of group composition and rule. Under unanimous rule, the effect of gender composition nearly disappears, and inequality nearly disappears with it, suggesting that composition and rule affect efficacy primarily by affecting speech participation. In other words, women's authority is negatively affected by unanimous rule primarily when they speak less than men; when women speak as much as men, they achieve equal authority. Under majority rule, the effect of composition also diminishes once we account for speech participation, with the exception of token women. In general, the analysis in Figure 6 shows that speech participation appears to be an important mediating mechanism for gender inequality.

Summary

We began the paper with the Gender Role hypothesis, and found that it received mixed support. Several findings support this hypothesis. First, we find that women are often disadvantaged in verbal participation, while men are never disadvantaged. Second, women participate less than their equal share when they are a minority and at equal rates when in a large majority (under some conditions; more on this below). Third, women do best in homogenous groups (and consistently across decision rule). Finally, women are more disadvantaged as tokens than they are in other compositions, and women tokens participate less than male tokens. Further support for gender role theory comes from disconfirmation of its competing hypotheses. Equal Disadvantage receives weak support at best. Violated Entitlement receives no support, as men never do worse than women in their group, not as even as tokens, and they do not talk less as tokens than they do as a minority in 2 male/3 female groups.³²

However, the Gender Role hypothesis fails to predict several other findings. First, in the majority decision-rule groups, it takes more to undo gender inequality than the hypothesis predicted – women must be a supermajority to achieve equality. Second and most importantly, opposite to the expectations prompted by gender role theory, in unanimous groups, women are not disadvantaged when in the minority. Finally, and most troublesome for the Gender Role hypothesis, female participation is far outstripped by male participation when women are a majority, just when they should be increasing their participation.

What can explain this set of findings is the interaction of decision rule and group gender composition: the Gender Role hypotheses about gender composition are largely correct under majority rule but largely incorrect under unanimous rule. The key point is that communication in deliberating groups exhibits gendered patterns, but the balance of voices is a result of the interaction between institutional factors, such as the group's decision rule, and the gender composition of deliberators. As predicted by our gender-rule interaction hypothesis, unanimous rule protects women in a numerical minority, and they take up their equal share of the conversation. Majority rule fails to protect women in the same way. Women in majority-rule groups tend to participate at much lower levels than the men in their groups (in terms of the percentage of the group conversation) and than

³² In addition, we found no evidence for the unique prediction of the Rooster hypothesis, which does not compete with gender role theory but also does not necessarily follow from it -- there is no special *benefit* from being a token male (relative to men in 2 male/3 female groups).

women in all-female groups (in raw talk times). Under the majority decision rule, only when women are a super-majority in their groups do they reach equality with men or with the levels of participation achieved by women in gender homogenous groups. These pervasive gender inequalities in the majority condition would seem to indicate that unanimity is the preferred decision rule. However, unanimous rule is a double-edged sword; while it protects minority status women, it protects minority status men even more. The reason seems to be that men leverage the protections of unanimous rule not to reach equality, as minority women do, but to exceed it. Thus, unanimous rule is no panacea; it can produce the highest gender inequality. However, unanimous rule is more robust than majority rule; it produces equality under more gender compositions.

These conclusions about rule would not be possible without interacting it with gender composition. Rule protects or hinders numerical minorities depending on who these minorities are.

Conclusion

Advocates of deliberative democracy posit equality of participation as a necessary requirement of deliberation. The results here show how far actual deliberation deviates from that ideal standard. Women speak substantially less than men in mixed-gender groups. Our findings show that speech is a crucial form of participation that substantially shapes perceptions of authority and influence. As critics of deliberation contend, deliberation can produce inequalities of participation that affect deliberators' sense of their own and others' efficacy. A key challenge for deliberation is how to create greater equality for women.

But these inequalities must be understood in light of the gender context and institutional rules within which men and women deliberate because the effect of gender is completely contingent on the structure of the group setting. This has direct implications for the debate among political theorists about whether deliberation is a positive force for democracy and its precept of equality or instead undermines the voices of subordinate group members. We suggest that a way to resolve this debate is to focus on the conditions that give rise to one or the other. Both views are empirically correct but contingent on circumstances. The fact that gender inequality disappears under some conditions means that deliberation can in fact meet the standard of equality, as its advocates contend.

Our results also have practical implications for policy making and efforts to reform democracy and enhance popular participation. It is possible to produce equal voice in deliberation through the adoption of simple rules such as the use of unanimous versus majority rule and by the assignment of deliberators to achieve particular gender compositions. Many government units and non-governmental organizations that use deliberation have the ability to shape the conditions of deliberation since they instigate, sponsor, and facilitate these discussions. Our results can be used as a set of instructions for people who already have the power and authority to shape conditions of discussion and wish to receive guidance on how to structure deliberative conversations in a more democratic direction.

The findings presented here provide several guidelines for promoting gender equality in a variety of deliberative settings ranging from local committees to citizen juries to civic groups. The participation of women equals that of men with respect to all our measures of verbal behavior either when women are outnumbered by men but receive the protection of unanimous decision rule, or

when women are the majority in groups that decide by majority rule. This means that if deliberations use majority rule to reach a decision, women need to be in the majority to exercise equal voice. It also means that if deliberations use unanimous rule, women need to be in a minority, so that they receive the protections of unanimity. Robust to decision rule are gender-homogenous groups. Men and women both do well in isolation from the other and decision rule then loses its power to undermine equality. But again, the key is to understand the interaction and use it for one's particular goal. If one wishes to avoid the maximum inequality then one should avoid groups with very few men and unanimous rule. If one wishes to minimize inequality of individual participation than one should construct groups with few women and unanimous rule. If one wishes to maximize the level of individual participation, one should construct gender homogenous groups and use either decision rule.

We offer caveats about the results. Our conclusions may not apply in particular circumstances. These include situations in which groups do not attempt to reach a collective decision; groups are strongly moderated; and when there are strong procedures or authoritative signals to speak equally.

More generally, this study reinforces the notion in several recent studies of political behavior that gender matters but does so contingent on the environment. Burns et al. (2001), for example, find that the gender composition of the political environment, whether operationalized as the density of women running for visible office in the voter's state, or the gender composition of civic groups, can significantly increase women's level of political information, interest, efficacy and participation. Such findings along with ours suggest the need for further work on ways in which gender composition and other gendered aspects of the environment might affect citizens (see also Crowder-Meyer 2010).

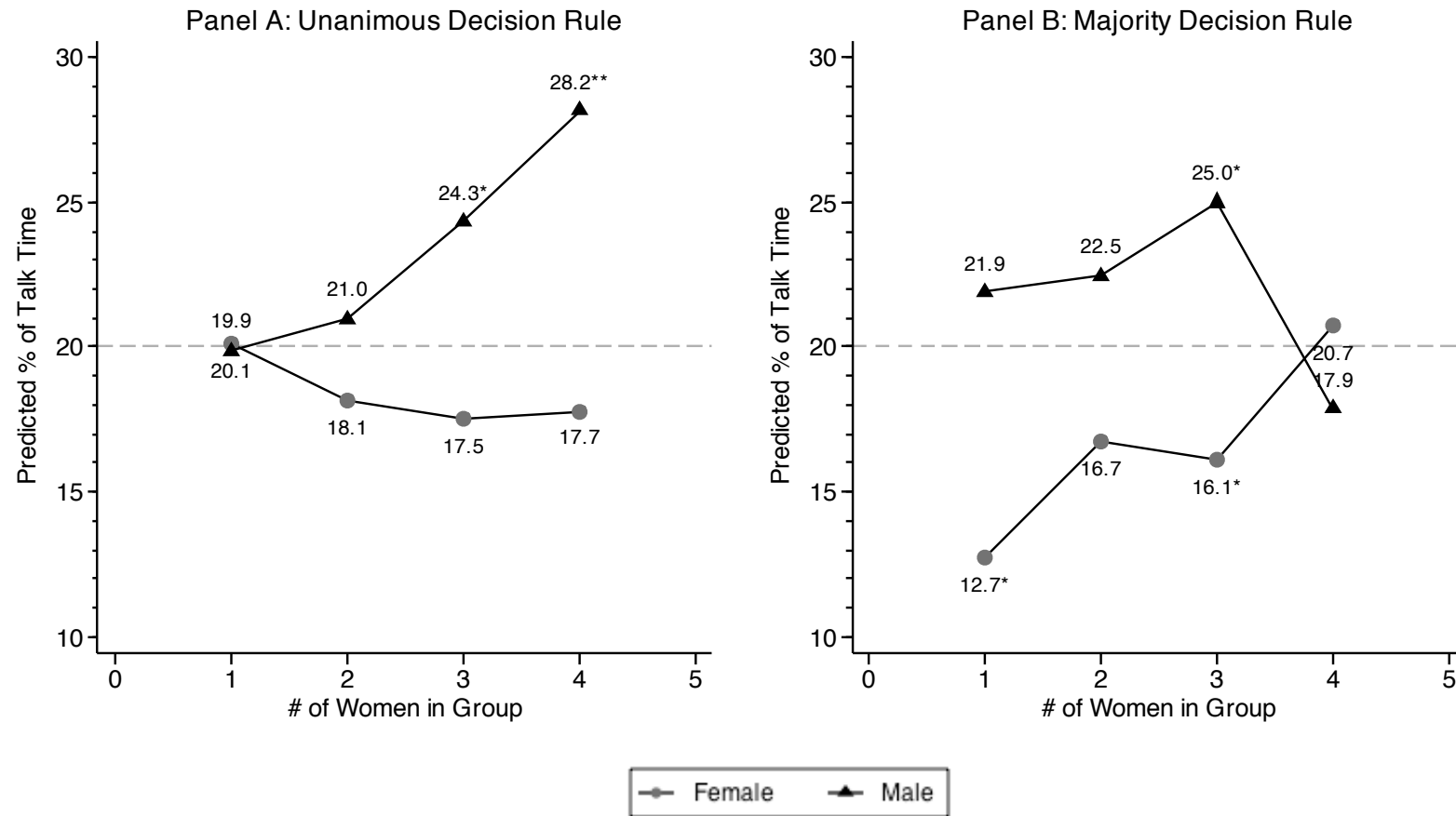
This implies a conceptual distinction between individual gender versus gender composition and other gendered aspects of settings and environments (see Sapiro 2003). An understanding of decision rules, for example, would be incomplete if it accounted for differences between men and women but not for differences across gendered compositions. Studies of political behavior or of political decision-making may not capture all that gender does if they only account for individual-level gender and omit gender composition and gendered interaction.

However, while the results we report here are in line with some existing hypotheses about gender composition, our chief result goes beyond the existing literatures. Political science has something unique to add to the study of gender relations: the notion that the institutional rules under which men and women participate in collective decision-making have a significant effect on gender dynamics. The interaction of gender composition and decision rule has not been a focus of any literature, yet we find that this interaction matters for both men and women.

Tables and Figures

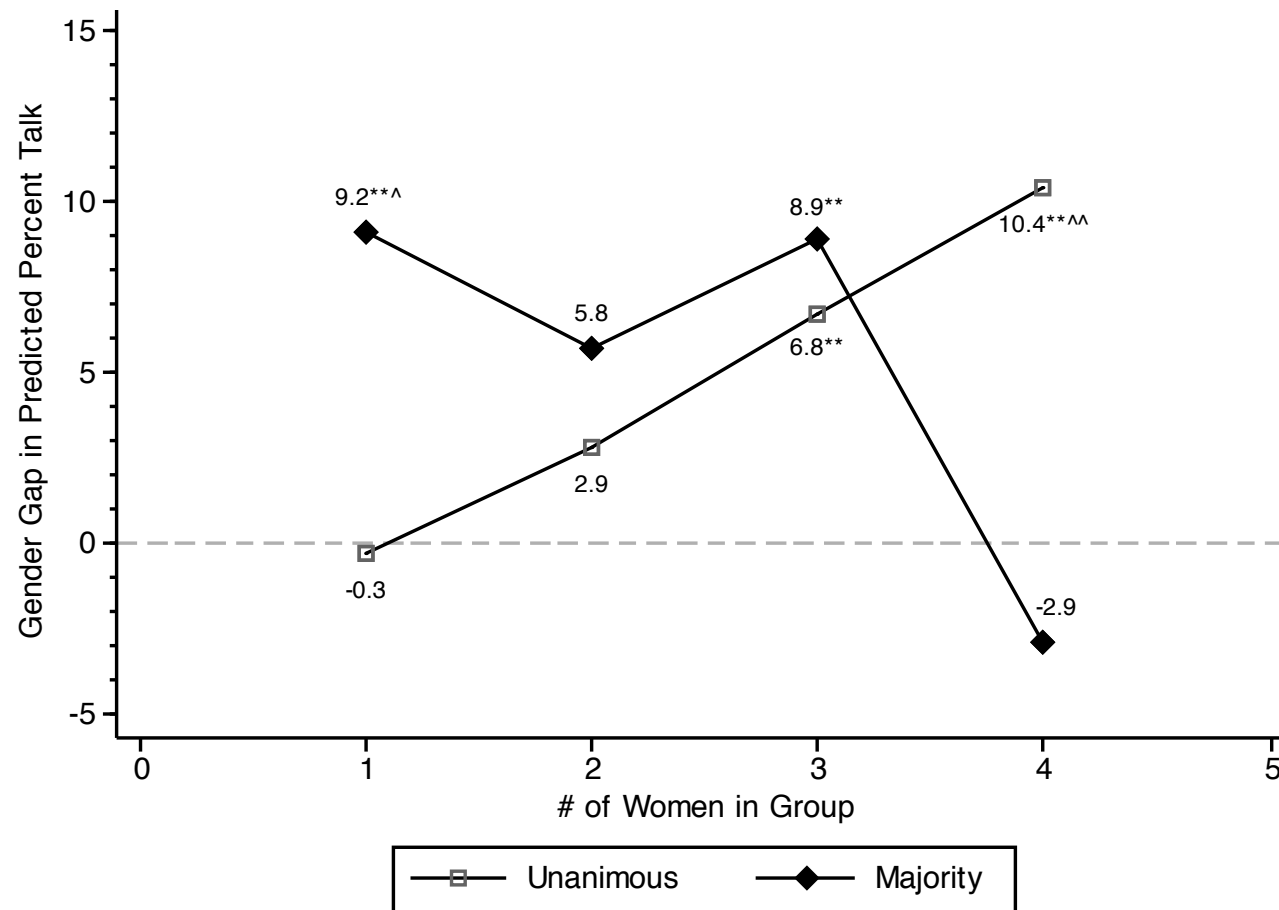
Table 1: Experimental Conditions and Ns				
	Unanimous	Majority	# of Groups	# of Individuals
0 Females	8	7	15	75
1 Female	9	9	18	90
2 Females	7	7	14	70
3 Females	9	7	16	80
4 Females	8	8	16	80
5 Females	8	7	15	75
# of Groups	49	45	94	
# of Individuals	245	225		470

Figure 1: Percent Talk by Experimental Condition with Controls for Outcome Preferences



Note: Stars indicate significant differences from 20% line of equality. ** $p < .05$, * $p < .10$, two-tailed test.

Figure 2: Gender Gaps Across Experimental Conditions



Note: Stars indicate gender gaps that are significantly different from 0. ** $p < .05$, * $p < .10$, two-tailed test. Carrots indicate significant differences in gender gap across decision rule. ^^ $p < .05$, ^ $p < .10$, one-tailed test.

Figure 3: Token Males and Females

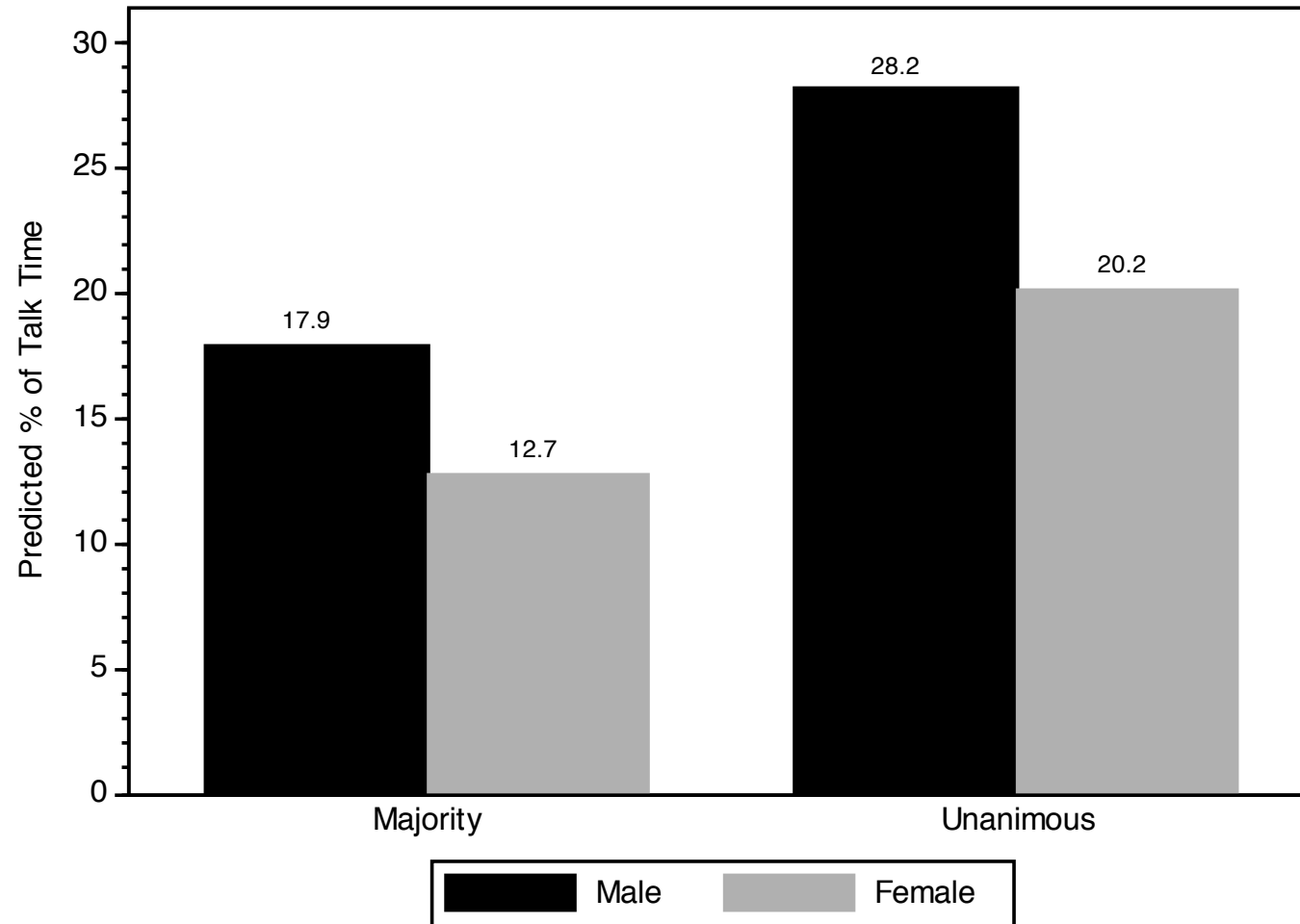
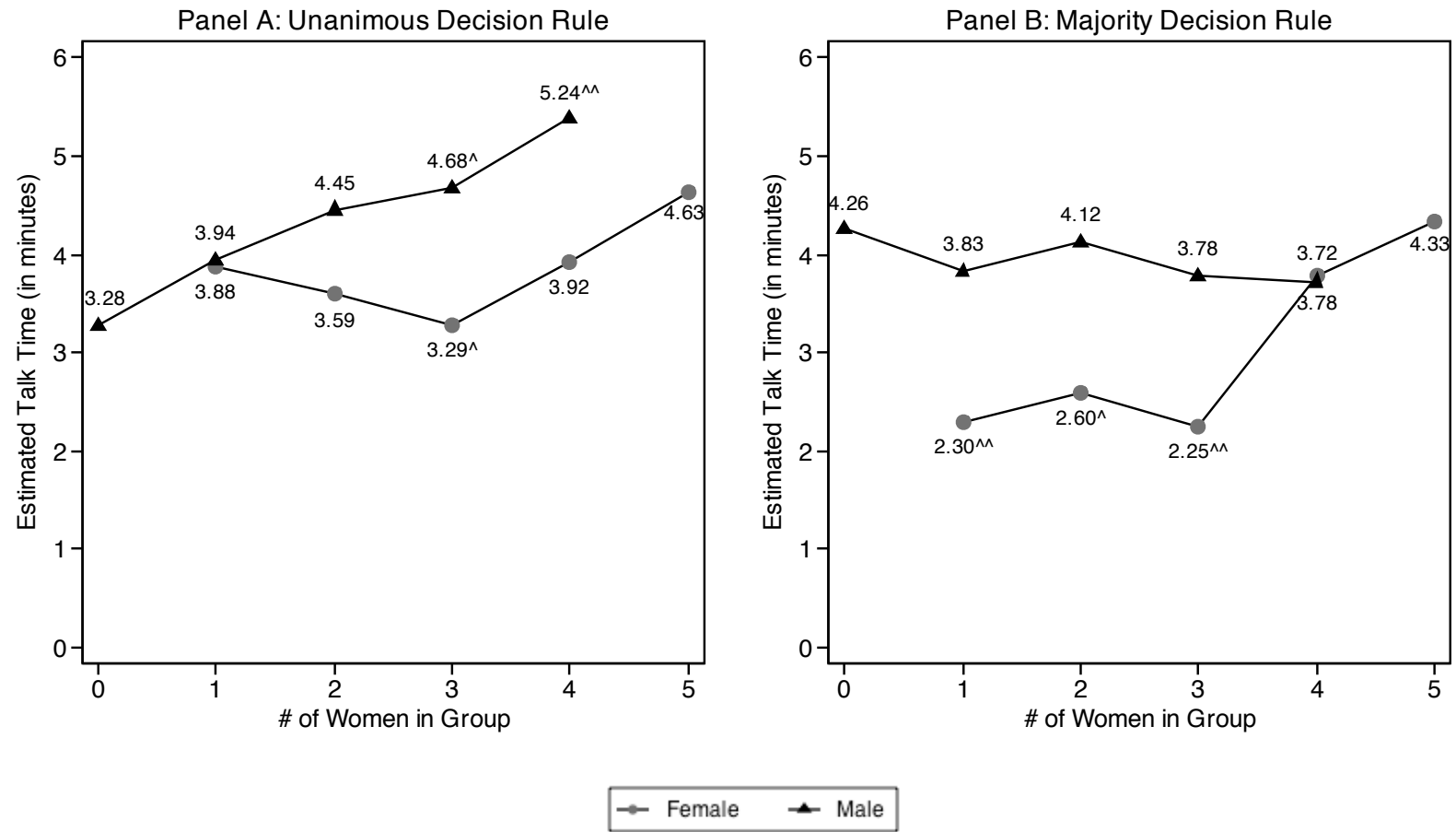


Figure 4: Total Talk Time by Experimental Condition



Note: Carrots indicate significant differences from the gender homogenous groups, using all-male groups as the comparison point for men, and all-female groups as the comparison point for women. ^^p.05, ^p<.10, one-tailed test.

Figure 5: Percent Talk, Self-Efficacy, and Other-Rated (Group) Efficacy

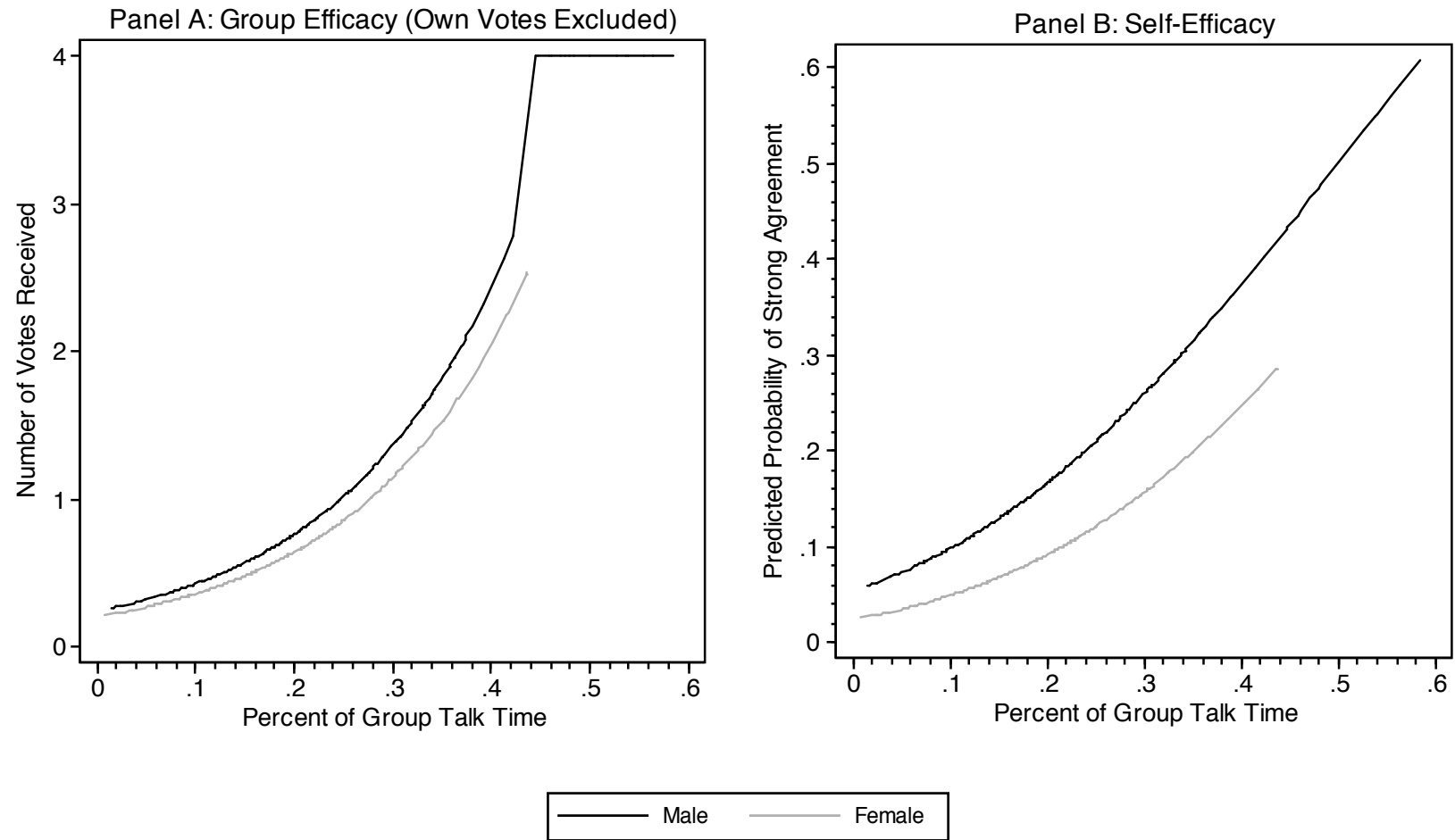
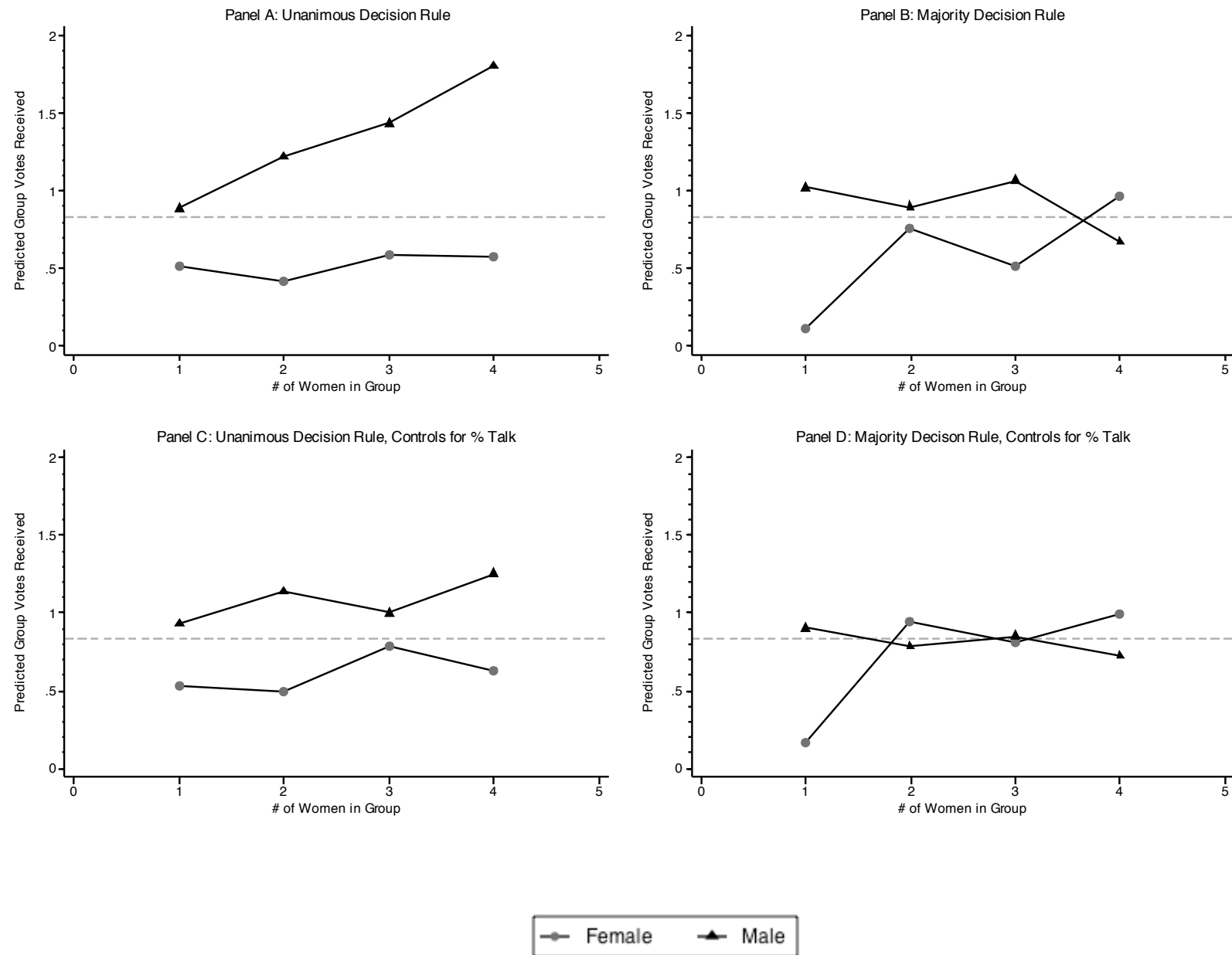


Figure 6: Predicted Group Votes Received, by Experimental Condition



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Table A2:
Determinants of Percent Speaking Turns by Gender Composition Condition

	1 Female/ 4 Males		2 Females/ 3 Males		3 Females/ 2 Males		4 Females/ 1 Male	
Female	-0.07	(0.03)	-0.04	(0.03)	-0.08	(0.03)	0.04	(0.03)
Unanimous	-0.02	(0.02)	-0.01	(0.03)	-0.02	(0.03)	0.10	(0.04)
Female*Unanimous	0.08	(0.05)	0.03	(0.05)	0.04	(0.04)	-0.13	(0.05)
Experimental Site	-0.01	(0.02)	-0.00	(0.02)	-0.00	(0.02)	0.00	(0.02)
Egalitarianism	0.05	(0.06)	-0.01	(0.07)	0.02	(0.06)	-0.04	(0.06)
Preference Matched Group Outcome	-0.03	(0.02)	-0.01	(0.02)	0.02	(0.02)	0.03	(0.02)
Intercept	.21	(0.03)	0.23	(0.04)	0.22	(0.04)	0.17	(0.05)
N	90		70		80		80	
Overall R ²	0.08		0.03		0.16		0.14	
<i>Coefficients generated from a random-effects regression, with random effects for each deliberating group. Standard errors are given in parentheses.</i>								

Table A3:
Determinants of Total Talk Time

	Men		Women	
Unanimous	-0.98	(0.93)	0.31	(0.95)
1 Female/4 Male Condition	-0.43	(0.92)	-2.02	(1.12)
1 Female*Unanimous	1.09	(1.28)	1.26	(1.60)
2 Female/3 Male Condition	-0.14	(1.02)	-1.72	(1.06)
2 Female*Unanimous	1.31	(1.43)	0.68	(1.53)
3 Female/2 Male Condition	-0.48	(1.10)	-2.08	(1.00)
3 Female*Unanimous	1.87	(1.49)	0.74	(1.39)
4 Female/1 Male Condition	-0.54	(1.26)	-0.54	(0.94)
4 Female*Unanimous	2.64	(1.77)	-0.17	(1.34)
Experimental Site	-2.98	(0.49)	-2.08	(0.48)
Egalitarianism	0.82	(1.11)	-0.79	(1.06)
Preference Matched Group Outcome	-0.47	(0.40)	0.25	(0.34)
Intercept	5.65	(0.88)	5.64	(0.89)
N (individuals)	237		233	
N (groups)	79		79	
Adj. R ²	0.22		0.21	
Coefficients generated from a random-effects regression, with random effects included for each deliberating group. Standard errors are given in parentheses.				

Table A4: Self-Evaluation of Efficacy	
Percent Talk	3.23 (0.57)
Confidence in ability to express opinions	0.54 (0.23)
Preference Matched Group Outcome	0.31 (0.11)
Female	-0.36 (0.11)
Experimental Site	-0.11 (0.09)
N	470
Pseudo R ²	.07
Log Pseudolikelihood	-498.01
<i>Coefficients generated from an ordered probit model. Standard errors are given in parentheses next to the coefficients. All standard errors are heteroskedasticity-robust and account for group clustering.</i>	

Table A5: Group Measures of Influence	
	Group Influence Votes (Own Vote Excluded)
Percent Talk	5.80 (0.49)
Confidence in ability to express opinions	0.32 (0.24)
Preference Matched Group Outcome	0.40 (0.11)
Female	-0.17 (0.11)
Experimental Site	-0.17 (0.06)
Intercept	-1.88 (0.23)
N	470
Log Pseudolikelihood	-515.20
<i>Coefficients generated from negative binomial regression. Standard errors are given in parentheses next to the coefficients. All standard errors are heteroskedasticity-robust and account for group clustering.</i>	

